JEPA-x: Cross-Predictive Physics Grounding for Forecastable Latent Dynamics

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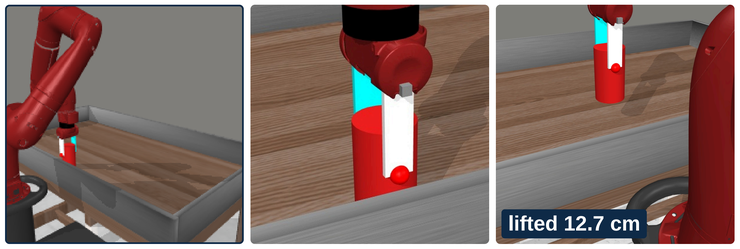
Human-centered Robotic Lab

1 Problem

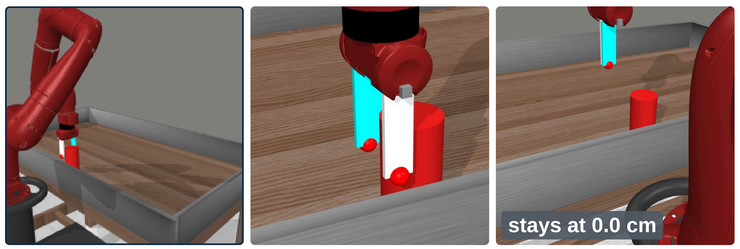
A visual world model must predict how a scene evolves from pixels alone. **Two scenes can look almost identical to the camera yet behave differently under the same action.**

Model's view 224 px input	**Actual scene** side view	**After the same lift**

Scene A · block held in the jaws



Scene B · block just behind the jaws



Illustrative scene. Model's views of A and B differ by 3.67 / 255 RMS per pixel.

Before the lift, the pixels barely differ, but the physical state (contact) already does. Self-prediction from pixels has little signal to separate A from B.

Idea

1 Alignment
Simulators give time-aligned **camera** and **physical-state** trajectories at training time.

2 Cross prediction
One shared predictor forecasts **both** futures from **either** view, so the visual latent must carry contact and pose.

$$\mathcal{L}_{\text{JEPA-}x} = \sum_{m,n \in \{o,s\}} \left\| g_\psi(Z_t^m, a_t) - z_{t+1}^n \right\|_2^2 \; + \; \beta \left[\Omega(z^o) + \Omega(z^s) \right]$$

$m = n$: within-modal (keeps LeWM self-prediction) · $m \neq n$: cross-modal · Ω: SIGReg isotropy regularizer, $\beta = 0.09$

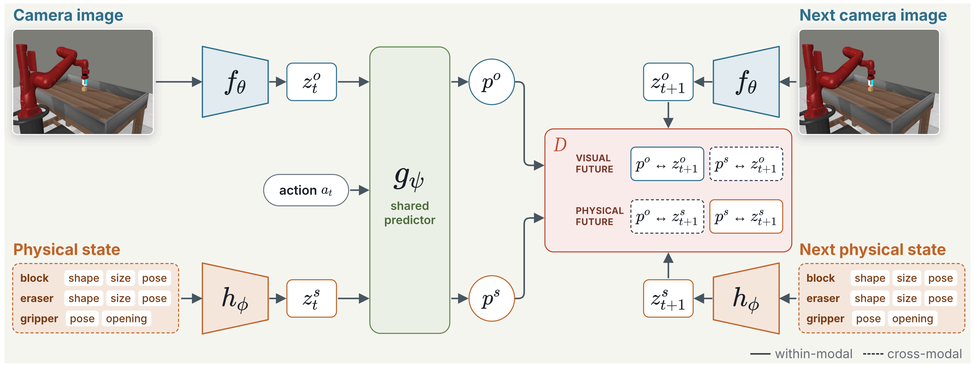
3 Visual-only deployment
The physical branch is removed at test time: pixels only, same planner as the baseline.

Takeaways

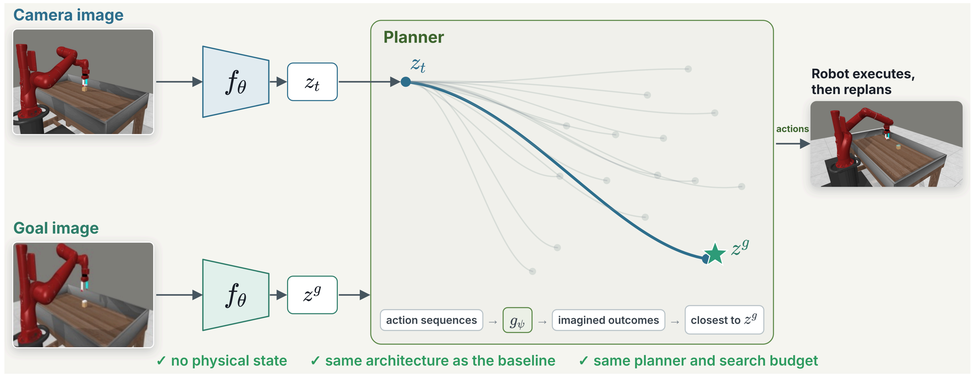
- Cross-modal prediction makes visual latents **more forecastable**, not just more decodable.
- Forecastable latents give **better planning from pixels**: 53.6 → 78.2% control.
- Physics is used **only in training**: same encoder, predictor and planner at test time.

2 Method

Training with paired trajectories



Deployment from pixels only



Planner: z-CEM over a 32-d action-VAE latent (5 macro-steps × 5 actions, 300 candidates, 30 iters), executes 25 actions, then replans. Same budget as the visual-only baseline LeWM (Maes et al., 2026).

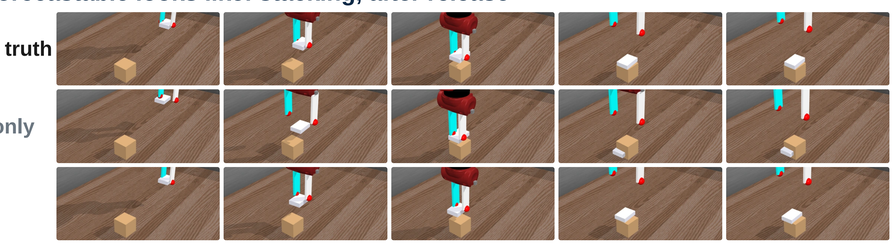
Decodable ≠ forecastable: cross-prediction is what improves control

Training objective	Decodable ↑ object-position R^2	Forecastable ↓ drift, new predictor	Useful ↑ control success %
Visual-only LeWM self-prediction	0.978	0.361	53.6
Regress + state regression	**0.991**	0.373	53.7
Cross-only separate predictors	–	0.101	73.0
JEPA-x shared predictor	**0.978**	**0.104**	**78.2**

Most decodable, no better control: regression makes position easiest to decode, yet the latent is no more forecastable. Cross-modal prediction is the main contributor; JEPA-x and Cross-only reach similar low drift.

Mean of 3 seeds. Distill: R^2 0.991, drift 0.516, control 57.7.

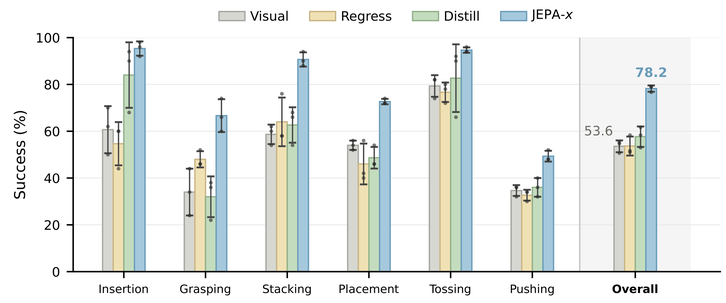
What forecastable looks like: stacking, after release



Decoded forecasts $t_0 \rightarrow t_4$ under identical actions (paper Fig. 5), not control executions. Visual-only predicts the eraser sliding off; JEPA-x keeps it supported.

3 Main Results

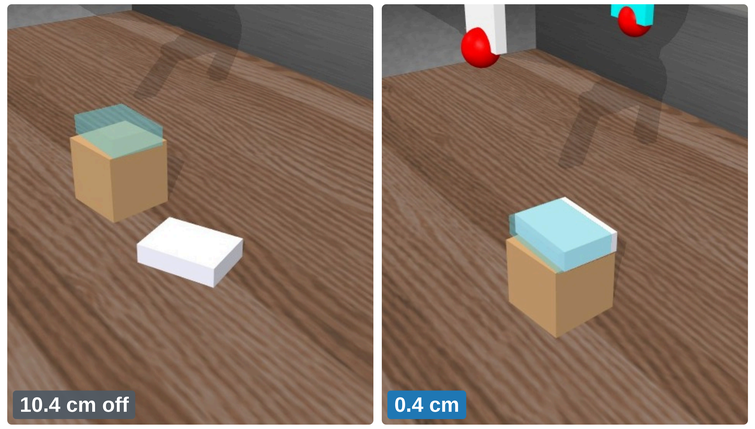
Control success per task family



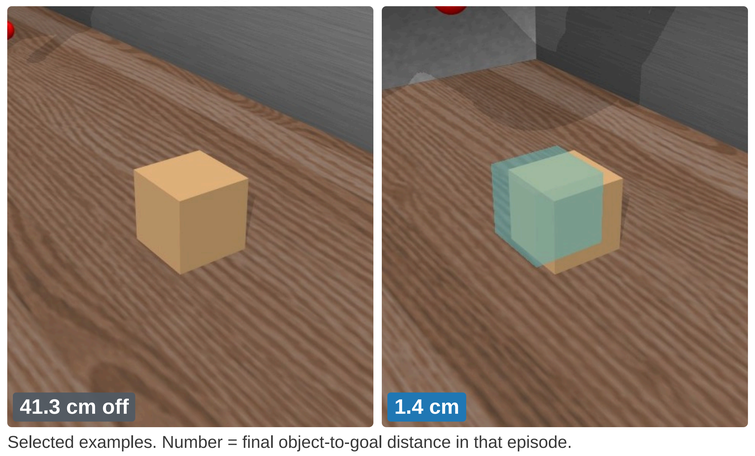
Same start, same goal

Visual-only | **JEPA-x**

Stacking eraser on block



Placement block to target



Selected examples. Number = final object-to-goal distance in that episode.

Paper
arxiv.org/abs/2608.24044

Website
wkh923.github.io/jepa-x